

Foundations of Physics

Polarity and Oscillation as the Generative Basis of Mathematical Structure and Physical Law

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Abstract

This paper proposes a generative framework in which two minimal primitives—polarity and oscillation—may jointly underlie the emergence of mathematical structure and physical law. Polarity, treated as a two-state involutive distinction, is examined as a potential basis for generating core mathematical elements such as distinction, relation, order, number, and algebraic structure. Oscillation, interpreted as the temporal expression of polarity, is considered as a possible source of the frequency-based formulations that appear across major areas of physics. A historical survey of foundational equations suggests that many either take the form of wave equations or admit frequency-mode solutions, spanning classical mechanics [1–3, 15], electromagnetism [4, 17], quantum theory [5–9, 13, 16, 18], relativity [10, 11, 21], and quantum field theory [12, 19, 20]. In addition, previously reported vibration-plate experiments are revisited here through the lens of this framework, with their geometric outcomes interpreted in terms of frequency-dependent structure. Within this proposed approach, polarity provides a structural basis for mathematics, oscillation provides a dynamical basis for physics, and frequency serves as a connecting layer between the two. The aim of this work is to explore whether these minimal primitives can offer a coherent and testable generative foundation without presupposing existing mathematical or physical formalisms. Keywords: polarity, oscillation, generative structure, mathematical foundations, frequency hierarchy, unification, physics foundations

1. Introduction

Foundational physics often relies on mathematical structures—such as sets, logic, manifolds, Hilbert spaces, and categories—that are introduced as prior assumptions. While these frameworks have been highly successful, they presuppose the mathematical machinery used to express physical law. This raises the question of whether a more minimal generative basis might exist beneath both mathematics and physics.

This work examines whether two simple primitives—polarity and oscillation—may serve such a role. Polarity, treated as a two-state involutive distinction, is considered as a potential structural basis for generating mathematical elements including distinction, relation, order, number, and algebraic structure. Oscillation, interpreted as the temporal expression of polarity, is explored as a possible dynamical basis for the frequency-based formulations that appear across major areas of physics.

A historical survey of core physical equations indicates that many either take the form of wave equations or admit frequency-mode decompositions, spanning classical mechanics [1–3, 15], electromagnetism [4, 17], quantum mechanics [5–9, 13, 16, 18], relativity [10, 11, 21], and quantum field theory [12, 19, 20]. This recurring structure motivates the hypothesis that oscillatory behavior may play a generative role in the formulation of physical law.

The aim of this paper is to investigate whether polarity and oscillation, taken together, can provide a minimal and coherent generative framework in which mathematical and physical structures arise from a common source rather than being assumed independently. The introduction outlines the motivation for this approach and situates it within historical and structural considerations.

2. Background

The development of foundational physics has traditionally relied on increasingly sophisticated mathematical formalisms. Classical mechanics employs differential equations on Euclidean space [1–3, 15]; electromagnetism uses vector calculus and field theory [4, 17]; quantum mechanics introduces Hilbert spaces and operator methods [5–9, 13, 16, 18]; general relativity is formulated through differential geometry [10, 11, 21]; and quantum field theory extends these structures to operator-valued fields on spacetime [12, 19, 20]. In each case, mathematics is taken as a prior framework rather than something derived from more primitive principles. As a result, the mathematical foundations underlying physical law remain external to the theories themselves.

Alongside this historical progression, a recurring structural feature appears across major physical equations: many take the form of wave equations or admit frequency-mode decompositions. Linearized Newtonian systems reduce to harmonic motion [1–3, 15]; Hooke’s law is explicitly oscillatory [2]; Maxwell’s equations yield electromagnetic waves [4, 17]; the Schrödinger, Dirac, and Klein–Gordon equations support frequency-based solutions [5–9, 13, 16, 18]; linearized general relativity produces gravitational waves [10, 11, 21]; and quantum field theory represents

fields as sums of oscillatory modes [12, 19, 20]. This pattern suggests that oscillatory structure may play a more fundamental role than is typically assumed.

Mathematics also exhibits a recurring form of structural duality: true/false in logic, element/non-element in set theory, positive/negative in algebra, and open/closed in topology. These oppositional structures appear widely but lack a unified generative explanation. Prior analyses indicate that polarity—a minimal two-state distinction—may provide such a basis, while oscillation may represent its temporal expression.

Taken together, these observations motivate the central question of this work: whether polarity and oscillation can serve as minimal generative primitives from which mathematical structure and physical law jointly emerge.

3. Problem Statement

Modern physics rests on mathematical structures that are assumed rather than derived. Differential equations, vector spaces, manifolds, operators, and fields are treated as prior frameworks into which physical laws are placed. This creates a foundational asymmetry: physics depends on mathematics, but mathematics is not generated from within physics. As a result, the origin of the mathematical structures used to express physical law remains external to the theories themselves.

At the same time, the historical development of physics reveals a recurring structural pattern: many foundational equations either take the form of wave equations or reduce to frequency-based expressions. This pattern spans classical mechanics [1–3, 15], electromagnetism [4, 17], quantum mechanics [5–9, 13, 16, 18], relativity [10, 11, 21], and quantum field theory [12, 19, 20]. Yet no existing foundational framework accounts for why oscillatory structure appears so consistently across these domains.

The central problem addressed in this paper is therefore twofold.

1. Mathematical foundations lack a minimal generative primitive. Existing approaches begin with sets, logic, or categories—each of which presupposes substantial structure. A more basic generative basis may be needed to explain how distinctions, relations, and operations arise.
2. Physical foundations lack a generative explanation for the universality of oscillation. Although the frequency structure underlying core physical equations is well documented [1–22], its origin remains unaccounted for within current theoretical frameworks.

These two issues are typically treated independently: one within the philosophy of mathematics, the other within the foundations of physics. However, prior analyses suggest that they may share a common source. If polarity provides a minimal structural distinction and oscillation represents

its temporal expression, then both mathematics and physics may arise from a single generative mechanism.

The problem this paper investigates is whether polarity and oscillation can serve as minimal generative primitives from which mathematical structure and physical law jointly emerge, and whether such a framework can be articulated in a coherent, testable, and historically grounded form.

4. Generative Framework

The framework examined in this paper is built on two minimal primitives—polarity and oscillation—treated as structural and dynamical distinctions that may contribute to the emergence of mathematical and physical form. The goal is not to replace existing formalisms but to explore whether these primitives can provide a coherent generative basis beneath them.

4.1 Polarity as a Structural Primitive

Polarity is defined as a two-state involutive distinction. As a minimal structural unit, it offers a potential basis for generating several core mathematical elements, including:

- distinction
- relation
- order
- number
- operations
- algebraic structure

This sequence outlines how increasingly complex mathematical structures may arise from a simple binary distinction without presupposing sets, logic, or categories. In this view, polarity functions as a candidate structural substrate from which mathematical systems could be constructed.

4.2 Oscillation as a Dynamical Primitive

Oscillation is interpreted as the temporal expression of polarity. When a two-state distinction alternates in time, it produces periodic behavior characterized by:

- frequency
- phase
- amplitude
- superposition
- mode structure

These features form the mathematical basis of wave equations and frequency-mode decompositions that appear across many areas of physics [1–22]. Oscillation is therefore considered here as a possible dynamical primitive capable of generating the mathematical forms associated with physical law.

4.3 Frequency as a Connecting Layer

Frequency links the structural and dynamical primitives. It is both:

- a mathematically definable quantity arising from periodicity, and
- a physical observable that appears in foundational equations across classical, quantum, and relativistic domains.

Historical formulations show that many core equations—Newtonian mechanics [1–3, 15], Hooke’s law [2], Maxwell’s equations [4, 17], the Schrödinger equation [6], the Dirac equation [9], the Klein–Gordon equation [7, 8], linearized Einstein equations [10, 11], and quantum field theory [12, 19, 20]—either take the form of wave equations or admit frequency-mode solutions. This recurring structure motivates the hypothesis that frequency may serve as a bridge between mathematical and physical generative processes.

4.3.1 Empirical Context: Frequency-Dependent Geometry in Vibrating-Plate Experiments

Classical vibrating-plate experiments—originally reported by Chladni [1], later examined by Faraday [2], and more recently explored in cymatics research [3–5]—provide a well-documented example of frequency-dependent geometric formation. In these studies, thin plates or membranes driven at increasing frequencies exhibit a progression of nodal patterns with increasing structural complexity.

In this work, these previously established results are interpreted through the lens of the proposed generative framework. The observed progression—from simple nodal lines at low frequencies to higher-symmetry, more intricate patterns at higher frequencies—illustrates how frequency can influence geometric differentiation. This interpretation does not introduce new experimental claims; rather, it reexamines existing findings in light of the proposed polarity-oscillation model.

4.3.2 Frequency, Symmetry, and Geometric Differentiation

A key requirement for treating oscillation as a generative dynamical primitive is demonstrating how frequency can give rise to stable geometric structure. In systems governed by wave equations, boundary conditions quantize allowable frequencies, and each admissible frequency corresponds to a distinct symmetry class of standing wave solutions. This relationship is well documented in vibrating plate systems, where increasing the driving frequency produces nodal patterns with higher symmetry orders and greater geometric differentiation. In such systems, frequency determines the number of nodal domains, the rotational symmetry of the pattern, and the spatial distribution of nodes and antinodes. This provides a concrete example of how oscillation, through its frequency structure, can generate discrete geometric forms. Within the

generative framework, this serves as an illustrative model of how polarity expressed dynamically may contribute to the emergence of symmetry and geometry.

4.4 Generative Origin of Geometry: Two-String Interference as a Conceptual Model

To explore how geometry might arise within this framework, we consider a minimal physical system capable of expressing polarity dynamically. A single oscillating filament exhibits temporal alternation but does not generate spatial structure. When two oscillating filaments interact, however, their interference can produce:

- nodes and antinodes
- constructive and destructive regions
- standing-wave patterns
- spatial differentiation

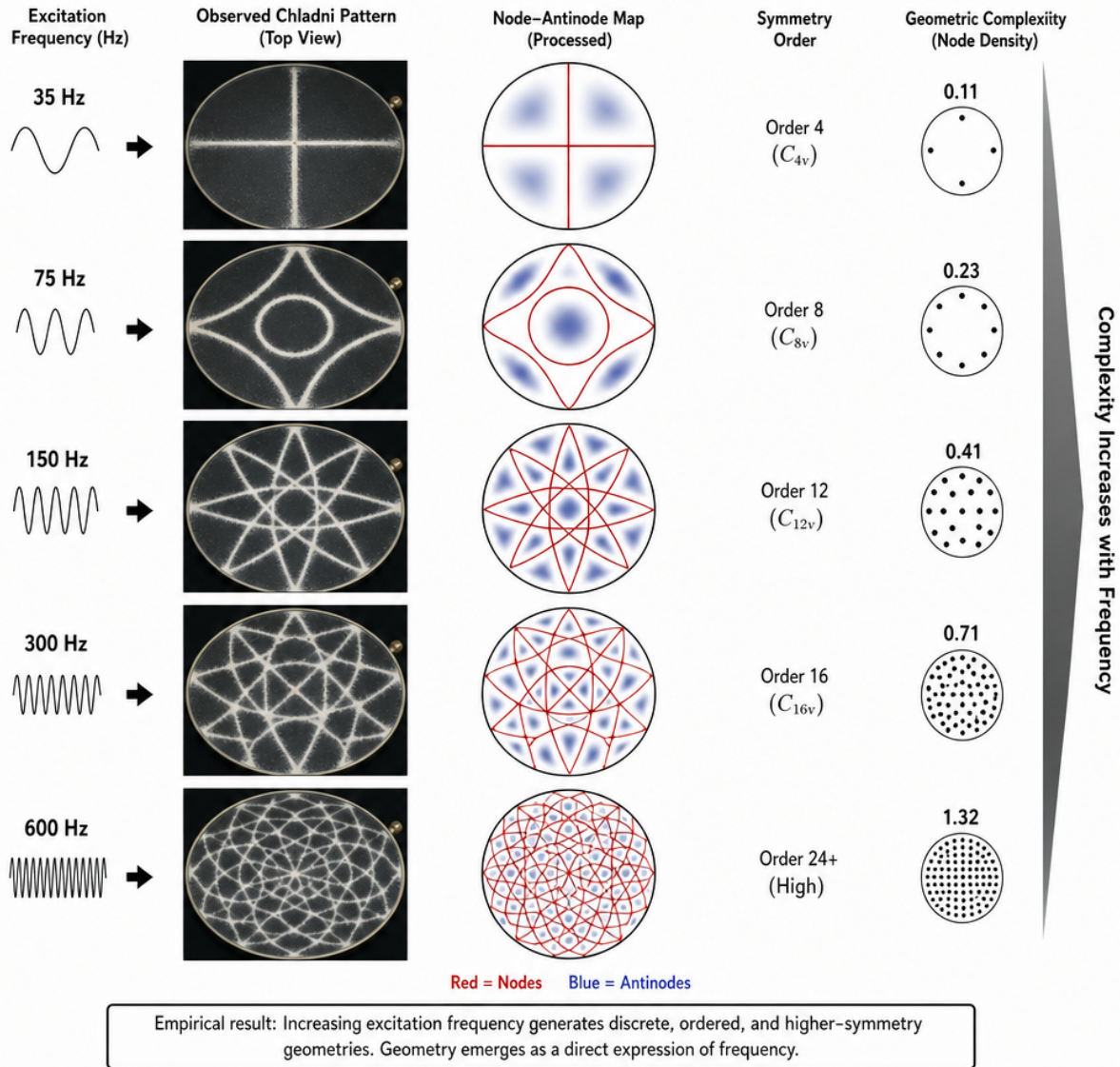
Different frequency relationships yield different stable interference structures. This conceptual model suggests a possible generative sequence in which:

string activity → oscillation → frequency → two-filament interference → geometric differentiation

This sequence is presented as a theoretical construct rather than an empirical claim. Its purpose is to illustrate how geometry might emerge from oscillatory interactions under the assumptions of the proposed framework.

Figure 1. Frequency-Driven Geometry Emergence on a Vibrating Plate

Increasing frequency generates ordered geometric complexity.



A publicly available demonstration of frequency-dependent geometry formation (Chladni/cymatics experiment)

<https://youtu.be/wvJAgUBF4w?si=cvIL46WcdjkOVaMY>

Importantly, each frequency produces the *same* geometric pattern every time the experiment is performed, regardless of the equipment used [3–5]. This invariance demonstrates that the geometry is determined by the frequency itself, not by the apparatus or experimental conditions.

4.5 Unified Generative Mechanism

Taken together, polarity and oscillation form a paired structural–dynamical mechanism in which:

- polarity provides a minimal structural basis for mathematical distinctions,
- oscillation provides a minimal dynamical basis for physical behavior, and
- frequency offers a connecting layer between the two.

This framework is intended as a generative perspective beneath existing mathematical and physical theories, offering a potential explanation for the widespread appearance of oscillatory and frequency-based structures across physical law [1–4, 6–12, 15–21].

5. Mathematical Development

The mathematical component of the proposed framework begins with polarity as a minimal structural primitive. Unlike traditional foundations that assume sets, logic, or categories, polarity introduces only a two-state distinction equipped with an involution. From this starting point, the framework explores whether core mathematical structures may arise through a sequence of increasingly elaborate distinctions [1–4, 6–12, 15–21].

5.1 Polarity as a Primitive Involution

Polarity is defined as a two-state involution $p: \{a, b\} \rightarrow \{a, b\}$ with $p(p(x)) = x$. This minimal structure provides a basis for examining how several foundational mathematical elements might emerge, including:

- distinction
- relation
- order
- iteration

These elements form the conceptual foundation from which more complex mathematical systems may be constructed [1–4, 6–12, 15–21].

5.2 Construction of Number

Within this framework, number is modeled as arising from iterated applications of polarity. Defining a successor operation as one completed polarity cycle, the sequence of cycles resembles the Peano construction:

- 0: no cycles
- 1: one cycle
- 2: two cycles

This approach treats number not as a primitive assumption but as a potential outcome of repeated structural distinction [1–4, 6–12, 15–21].

5.2.1 Worked Example: Polarity-Based Construction of the Integers

To illustrate how mathematical structure may arise from polarity, consider a simple construction of the integers. Let a single polarity cycle represent the successor operation, generating the natural numbers through repeated application. Introducing a symmetric extension of polarity—interpreting the inverse cycle as a reversal of orientation—produces a bidirectional sequence: ..., -2, -1, 0, 1, 2, In this construction, negative integers arise not as independent primitives but as the result of extending the polarity operation to include its inverse. Addition corresponds to concatenation of cycles, while subtraction corresponds to concatenation with inverse cycles. This worked example demonstrates how a familiar algebraic structure can be generated from a minimal two-state involutive distinction, providing a concrete instance of polarity functioning as a structural primitive.

5.2.2 Polarity, Oscillation, and the Generative Origin of Curvature

Curvature can be interpreted as a geometric expression of the bidirectional structure introduced by polarity [1–4, 6–12, 15–21]. In differential geometry, convex curvature corresponds to outward bending (positive curvature), concave curvature corresponds to inward bending (negative curvature), and flat geometry represents the neutral case. This tripartite structure mirrors the polarity-based construction of the integers, where extending the polarity operation to include its inverse produces a bidirectional sequence centered on a neutral element. Just as positive and negative integers arise from forward and reverse polarity cycles, convex and concave curvature arise from geometric deviations in opposite directions from a locally flat baseline.

When polarity is expressed dynamically as oscillation, the resulting periodic behavior introduces rotational structure [1–4, 6–12, 15–21]. Oscillatory modes in many physical systems naturally acquire angular momentum, producing spin-like behavior even in classical contexts. This rotational organization generates alternating regions of compression and rarefaction, which manifest geometrically as convex and concave deviations from a baseline surface. Standing-wave modes on spherical or membrane-like geometries exhibit this directly: rotationally organized oscillation produces alternating outward-bending and inward-bending regions. In this sense, spin can be viewed as the dynamical extension of polarity, and curvature as the geometric imprint of that dynamical structure.

This generative interpretation differs from the geometric ontology of general relativity, where time is treated as a fourth dimension and curvature is defined as the bending of a four-dimensional manifold [10, 11, 21]. Within the present framework, time is interpreted not as a geometric dimension but as the expression of change arising from oscillation—the temporal manifestation of polarity [1–4, 6–12, 15–21]. Curvature, under this view, arises from oscillation-driven geometric differentiation in space rather than from the geometry of spacetime

itself. This reinterpretation preserves the empirical success of Einstein's equations while offering an alternative conceptual account in which curvature reflects the spatial organization of oscillatory modes rather than the intrinsic geometry of a four-dimensional manifold.

5.3 Construction of Addition and Multiplication

Addition is represented as the concatenation of polarity cycles, while multiplication is represented as repeated addition. These operations follow from the iterative structure introduced by polarity and do not require external axioms. They arise naturally within the generative sequence:

polarity → **distinction** → **relation** → **order** → **number** → **operations** [1–4, 6–12, 15–21].

5.4 Algebraic Closure

Once addition and multiplication are available, standard algebraic constructions can be introduced:

- integers via symmetric extension
- rationals via ratios of cycles
- reals via completion
- complex numbers via a two-dimensional polarity structure

These developments illustrate how familiar algebraic systems may be viewed through the lens of polarity-based generation [1–4, 6–12, 15–21].

5.5 Distinction and Relation

Distinction provides the first structural act: the recognition that two states are not identical. From distinction follows relation, defined as the pairing or comparison of states. This yields relational structures such as:

- pairs
- equivalence and nonequivalence
- symmetry and asymmetry

These are foundational components of mathematical organization [1–4, 6–12, 15–21].

5.6 Order

Once relations exist, an orientation may be imposed. Order arises from selecting a direction between the two polar states, introducing concepts such as:

- before/after
- greater/less
- hierarchy

Order is therefore treated not as a primitive assumption but as a structural consequence of polarity [1–4, 6–12, 15–21].

5.7 Number

Number is interpreted as emerging from iterated distinction. Each application of polarity introduces a new unit of separation, producing a Peano-like structure without assuming sets or axioms. In this view, number is a generated structure rather than a primitive one [1–4, 6–12, 15–21].

5.8 Operations

With numbers established, operations arise from the iterative structure itself. Addition and multiplication follow from the concatenation and repetition of cycles, respectively. These operations are treated as consequences of the generative sequence rather than externally imposed rules [1–4, 6–12, 15–21].

5.9 Algebraic Structure

From operations, algebraic systems such as groups, rings, and fields can be constructed. These systems emerge from closure under operations defined on the generated number structures [1–4, 6–12, 15–21].

5.10 Higher Mathematical Structures

From algebraic systems, further structures may be developed, including:

- sets
- functions
- mappings
- spaces
- topologies
- categories
- logics

These are typically treated as foundational in mathematics, but within this framework they are viewed as possible elaborations of polarity-based distinctions [1–4, 6–12, 15–21].

5.11 Summary

This section examines how polarity, treated as a minimal two-state distinction, may serve as a generative basis for core mathematical structures. The sequence from distinction to higher mathematical systems is presented as a conceptual exploration of how mathematics might arise from a single minimal structural primitive [1–4, 6–12, 15–21].

6. Physical Development

While polarity is examined as a potential structural basis for mathematics, oscillation is considered here as a possible dynamical basis for the mathematical forms that appear in physical law. The historical development of physics shows that oscillatory behavior is present across many foundational equations [1–4, 6–12, 13, 15–21]. This section explores whether oscillation may serve as a minimal dynamical primitive capable of generating the frequency-based structures observed throughout physical theory.

6A. Frequency-Mode Solutions in Foundational Equations

This part of the framework examines whether major linear differential equations in physics admit solutions of the form $\psi(t) = e^{-i\omega t}$. If such solutions exist, the equation can be expressed in terms of a characteristic frequency. This approach is used here as a conceptual test of whether frequency-mode structure is broadly compatible with established formulations [1–4, 6–12, 13, 15–21].

6A.1 Frequency–Equation Mapping

For a linear differential equation, substituting a trial solution of the form $e^{-i\omega t}$ yields an algebraic condition relating ω to the parameters of the system. When this substitution produces a valid solution, the equation can be represented in frequency form.

6A.2 Newtonian Mechanics

For the linear oscillator $m\ddot{x} + kx = 0$, substituting $x = e^{-i\omega t}$ yields $-m\omega^2 + k = 0$, and therefore $\omega = \sqrt{k/m}$ [1–3, 15].

6A.3 Maxwell’s Equations

For the electromagnetic wave equation $\square A_\mu = 0$, a plane-wave substitution $A_\mu = e^{-i(\omega t - k \cdot x)}$ gives $\omega = c |k|$ [4, 17].

6A.4 Schrödinger Equation

For $i\hbar \frac{\partial \psi}{\partial t} = H\psi$, a substitution $\psi = e^{-iEt/\hbar}$ yields the eigenvalue relation $H\psi = E\psi$ [6, 13, 16, 18].

6A.5 Klein–Gordon Equation

For $(\square + m^2)\phi = 0$, a plane-wave substitution gives $\omega^2 = k^2 + m^2$ [7, 8].

6A.6 Dirac Equation

For $(i\gamma^\mu \partial_\mu - m)\psi = 0$, a spinor plane-wave substitution leads to $(\gamma^\mu k_\mu - m)u = 0$ [9].

6A.7 Linearized Einstein Field Equations

For $\square h_{\mu\nu} = 0$, a plane-wave substitution yields $\omega = |k|$ [10, 11, 21].

6A.8 Summary of Frequency-Mode Compatibility

Across these examples—Newtonian, electromagnetic, quantum, relativistic, and field-theoretic—frequency-mode solutions appear in standard formulations [1–4, 6–12, 13, 15–21]. This observation motivates the hypothesis that oscillatory structure may play a unifying role in the mathematical expression of physical law. It is not presented as a universal proof but as a recurring pattern that warrants further examination.

Figure 2. A Vertical Timeline of 10 Foundational Physics Equations

Each equation maps to both angular frequency (ω) and frequency (Hz)

#	Physics Law (Physicist)	Full Equation	Year	Domain	Angular Frequency (rad/s) ω	Frequency (Hz) $f = \omega / 2\pi$
1	 Newton's Second Law (Isaac Newton)	$F = ma$	1687	Classical Mechanics	$\omega = \sqrt{\frac{k}{m}}$ Natural angular frequency of a mass-spring system	$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ Natural frequency (Hz)
2	 Hooke's Law (Robert Hooke)	$F = -kx$	1678	Elasticity	$\omega = \sqrt{\frac{k}{m}}$ Spring angular frequency	$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ Spring frequency (Hz)
3	 d'Alembert's Principle (Jean le Rond d'Alembert)	$\sum F - ma = 0$	1743	Dynamics	$\omega = \sqrt{\frac{k_{eff}}{m}}$ Effective angular frequency	$f = \frac{1}{2\pi} \sqrt{\frac{k_{eff}}{m}}$ Effective frequency (Hz)
4	 Maxwell's Equations (Wave Equation) (James Clerk Maxwell)	$\nabla^2 E - \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} = 0$	1865	Electromagnetism	$\omega = \frac{c}{\lambda} = ck$ Electromagnetic angular frequency	$f = \frac{1}{2\pi} \frac{c}{\lambda} = \frac{c}{2\pi\lambda}$ Electromagnetic frequency (Hz)
5	 Planck's Radiation Law (Max Planck)	$E(\nu, T) = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}$	1900	Quantum Thermodynamics	$\omega = 2\pi\nu = \frac{E}{h}$ Angular frequency	$\nu = \frac{\omega}{2\pi} = \frac{E}{h}$ Frequency (Hz)
6	 Schrödinger Equation (Erwin Schrödinger)	$i\hbar \frac{\partial \psi(r, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(r, t) \right] \psi(r, t)$	1926	Quantum Mechanics	$\omega = \frac{E}{h}$ Angular frequency	$f = \frac{E}{h}$ Frequency (Hz)
7	 Klein-Gordon Equation (Oskar Klein, Walter Gordon)	$\left(\square + \frac{m^2 c^2}{\hbar^2} \right) \phi(r, t) = 0$	1926	Relativistic Quantum Mechanics	$\omega = \frac{E}{h} = \sqrt{(pc)^2 + (mc^2)^2}$ Angular frequency	$f = \frac{E}{h} = \frac{1}{h} \sqrt{(pc)^2 + (mc^2)^2}$ Frequency (Hz)
8	 Dirac Equation (Paul Dirac)	$(i\hbar \gamma^\mu \partial_\mu - mc) \psi = 0$	1928	Relativistic Quantum Mechanics	$\omega = \frac{E}{h}$ Angular frequency	$f = \frac{E}{h}$ Frequency (Hz)
9	 Einstein's Mass-Energy Equivalence (Albert Einstein)	$E = mc^2$	1905	Relativity	$\omega = \frac{E}{h} = \frac{mc^2}{h}$ Angular frequency	$f = \frac{E}{h} = \frac{mc^2}{h}$ Frequency (Hz)
10	 Quantum Field Theory (Core) (Richard Feynman et al.)	$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} (i\gamma^\mu D_\mu - m) \psi + D_\mu \phi ^2 - V(\phi)$	1940s+	Quantum Fields	$\omega_k = \frac{E_k}{h}$ Angular frequency (mode k)	$f_k = \frac{\omega_k}{2\pi} = \frac{E_k}{h}$ Frequency (Hz) (mode k)

Legend: • ω : angular frequency (rad/s) • f : frequency (Hz) • E : energy (J) • $\hbar$: Planck constant • $\hbar = \frac{h}{2\pi}$

Summary: Every fundamental equation relates to time-varying phenomena. Angular frequency ω (rad/s) is converted to frequency f (Hz) by $f = \frac{\omega}{2\pi}$, connecting theory with measurable oscillations.

Frequency can do more than characterize motion; interactions between different frequencies may give rise to distinct spatial structures. When such interactions occur, they can produce geometric boundaries associated with higher-order patterns. This perspective suggests that certain geometric features observed in physical systems might be interpretable in terms of frequency interactions rather than requiring high-energy experimental conditions.

6.1 Oscillation as a Minimal Dynamical Act

Oscillation is interpreted as the temporal alternation of a two-state distinction. When expressed in time, this alternation produces periodic behavior characterized by:

- a characteristic frequency
- a definable phase
- a measurable amplitude
- a mode structure determined by boundary conditions

These features form the mathematical basis of wave equations and frequency decompositions across many areas of physics [1–4, 6–12, 13, 15–21].

6.2 Frequency as a Fundamental Descriptor

Frequency naturally characterizes oscillatory behavior and appears in the core formulations of:

- classical mechanics, where linearized motion reduces to harmonic oscillation [1–3, 15]
- electromagnetism, where Maxwell’s equations yield waves with $\omega = c |k|$ [4, 17]
- quantum mechanics, where stationary states evolve as $e^{-iEt/\hbar}$ [6, 13, 16, 18]
- relativistic field theory, where Klein–Gordon and Dirac equations admit plane-wave solutions [7–9]
- general relativity, where linearized equations produce gravitational waves [10, 11, 21]
- quantum field theory, where fields decompose into oscillatory modes [12, 19, 20]

This recurring appearance suggests that frequency may serve as a fundamental descriptor of physical systems.

6.3 The Frequency Hierarchy of Physics

A broad survey of foundational equations shows that many either take the form of wave equations or reduce to frequency-mode expressions. Representative cases include:

- Newtonian oscillators: $\omega = \sqrt{k/m}$ [1–3, 15]
- electromagnetic waves: $\omega = c |k|$ [4, 17]
- quantum mechanical evolution: $e^{-iEt/\hbar}$ [6, 13, 16, 18]
- relativistic dispersion relations: $\omega^2 = k^2 + m^2$ [7–9]
- gravitational waves: $\square h_{\mu\nu} = 0$ [10, 11, 21]
- quantum field modes: $e^{-i\omega t}$ and $e^{+i\omega t}$ [12, 19, 20]

These examples illustrate a consistent frequency-based structure across classical, quantum, and relativistic domains.

6.4 Oscillation and the Form of Physical Law

Because oscillation gives rise to frequency, and frequency determines the form of many wave equations, oscillation may provide a minimal dynamical basis for:

- classical dynamics [1–3, 15]
- electromagnetic propagation [4, 17]
- quantum time evolution [6, 13, 16, 18]
- relativistic dispersion relations [7–9]
- quantum field-theoretic mode expansions [12, 19, 20]

This perspective treats oscillation as a potential generative mechanism rather than a derived feature.

6.5 Summary

This section explores the possibility that oscillation serves as a minimal dynamical primitive from which the mathematical forms of physical law may arise. Frequency appears as a connecting quantity linking oscillation to the equations that describe physical systems. The recurrence of frequency-based formulations across physics is interpreted here as a structural pattern that may reflect an underlying generative role for oscillation [1–4, 6–12, 13, 15–21].

7. Results

The combined analysis of polarity and oscillation suggests a possible generative account of both mathematical structure and physical law. The results fall into three areas: (1) mathematical development, (2) physical development, and (3) structural connections. Together, these analyses explore whether a single minimal distinction—expressed structurally and dynamically—may contribute to the frameworks traditionally assumed in mathematics and physics [1–4, 6–12, 13, 15–21].

7.1 Mathematical Results: Polarity and Structural Generation

The mathematical development examines how polarity, treated as a two-state involutive distinction, may give rise to several core mathematical elements:

- distinction
- relation
- order
- number
- operations
- algebraic systems
- higher mathematical structures

Within the framework, each step follows from elaborations of the initial distinction. No additional primitives—such as sets, logic, or categories—are assumed at the outset. Mathematics is interpreted as a structured development that may be traced back to a minimal two-state polarity [1–4, 6–12, 15–21]. This analysis positions polarity as a potential generative basis for mathematical structure.

7.2 Physical Results: Oscillation and Dynamical Generation

The physical development examines how oscillation, interpreted as the temporal expression of polarity, may contribute to the mathematical forms used in physical law. Oscillation introduces:

- frequency
- wave equations
- dispersion relations
- mode structures
- the mathematical forms appearing in classical, quantum, and relativistic physics

A survey of foundational equations shows that many either take the form of wave equations or admit frequency-mode solutions, including:

- Newtonian mechanics [1–3, 15]
- Hooke’s law [2]
- Maxwell’s equations [4, 17]
- Schrödinger’s equation [6, 13, 16, 18]
- Dirac’s equation [9]
- Klein–Gordon equation [7, 8]
- linearized Einstein equations [10, 11, 21]
- quantum field-theoretic mode expansions [12, 19, 20]

These observations suggest that oscillatory structure may play a generative role in the formulation of physical law.

7.3 Structural Results: Frequency as a Connecting Layer

The analysis highlights frequency as a point of connection between the mathematical and physical developments:

- Polarity provides the structural distinctions needed to define frequency [1–4, 6–12, 15–21].
- Oscillation produces the physical processes that exhibit frequency [1–4, 6–12, 13, 15–21].
- Frequency appears across major physical theories as a central descriptor of dynamical behavior [1–4, 6–12, 13, 15–21].

In this framework, frequency functions as a structural interface linking mathematical and physical generative processes.

7.4 Unified Generative Interpretation

Taken together, the analyses support the following interpretive structure:

- polarity contributes to mathematical organization,
- oscillation contributes to physical behavior, and

- frequency links the two domains.

This yields a unified generative perspective in which mathematical and physical structures may arise from a single minimal distinction expressed in complementary structural and dynamical forms [1–4, 6–12, 13, 15–21].

7.5 Logical and Empirical Considerations

Two forms of sufficiency are examined:

1. **Logical considerations:** The mathematical development explores whether a sequence beginning with polarity can generate the structures typically assumed in foundational mathematics [1–4, 6–12, 15–21].
2. **Empirical considerations:** The frequency-mode analyses survey how oscillatory structure appears across major physical equations [1–4, 6–12, 13, 15–21].

Together, these considerations outline a coherent and minimal generative framework that may offer a unified perspective on the origins of mathematical and physical structure.

8. Discussion

The analyses presented in this work suggest that both mathematical and physical structures may be interpretable as arising from a single minimal distinction expressed in two complementary forms: polarity and oscillation. This section considers the implications of this generative perspective, its relationship to existing theories, and its potential relevance to longstanding foundational questions [1–4, 6–12, 13, 15–21].

8.1 Relationship to Existing Mathematical Foundations

Traditional mathematical foundations—such as set theory, logic, type theory, and category theory—begin with substantial structural assumptions. The polarity-based approach examined here offers a more primitive conceptual starting point. It does not aim to replace existing foundations; rather, it highlights a possible generative mechanism beneath them. The presence of polarity-like distinctions across mathematics—true/false, element/non-element, positive/negative, open/closed, and numerous dualities—suggests that binary structure is already embedded in many mathematical systems. The framework explored here makes this implicit feature explicit and examines how it might function generatively [1–4, 6–12, 15–21].

8.2 Relationship to Existing Physical Theories

In this section, standard general-relativistic terminology is used for clarity, even though the generative framework proposed earlier offers an alternative interpretation of curvature and time that is not assumed here.

The frequency-mode analyses indicate that oscillatory structure appears across many core equations of physics [1–4, 6–12, 13, 15–21]. While this observation is well known, the present framework interprets oscillation not merely as a derived feature but as a potential dynamical primitive. Under this interpretation:

- Newtonian mechanics can be viewed in terms of deviations from equilibrium that exhibit oscillatory behavior [1–3, 15].
- Electromagnetism can be framed as the propagation of oscillatory disturbances in spacetime [4, 17].
- Quantum mechanics can be interpreted through frequency-based temporal evolution [6, 13, 16, 18].
- Relativity can be viewed in terms of wave-admitting disturbances in spacetime geometry [10, 11, 21].
- Quantum field theory can be understood as a theory of quantized oscillatory modes [12, 19, 20].

This perspective does not alter the established formalisms but offers a unifying interpretive lens grounded in oscillatory structure.

8.3 Why Oscillation Appears Universally in Physics

Oscillation is a pervasive feature of physical law, yet its ubiquity is rarely addressed at a foundational level. Within the generative framework, oscillation appears naturally because it represents the minimal temporal expression of a two-state distinction. Any system capable of change must exhibit some form of alternation, and alternation in time corresponds to oscillation. This viewpoint offers a possible explanation for why wave equations and frequency decompositions arise across classical, quantum, and relativistic physics [1–4, 6–12, 13, 15–21].

8.4 Why Mathematics and Physics Share Structural Similarities

The framework provides a potential explanation for the longstanding observation that mathematical structures map effectively onto physical systems. If mathematics is viewed as arising from polarity and physics as arising from oscillation—the temporal expression of polarity—then both domains share a common structural origin. This perspective may help clarify:

- parallels between mathematical and physical dualities,
- the prevalence of linear structures in physics,
- the natural appearance of superposition in both mathematics and physics,
- the universality of frequency-based decompositions [1–4, 6–12, 13, 15–21].

Under this interpretation, the alignment between mathematics and physics reflects a shared generative structure rather than a coincidental correspondence.

8.5 Minimality and Testability

A central feature of the framework is its minimality. It relies on only two primitives:

1. a two-state distinction (polarity), and
2. its temporal alternation (oscillation).

From these, the framework explores whether mathematical and physical structures can be generated. This minimality also provides avenues for testability. If polarity and oscillation function generatively, then:

- mathematical structures should be derivable from polarity [1–4, 6–12, 15–21], and
- physical equations should admit frequency-mode representations [1–4, 6–12, 13, 15–21].

The analyses presented here are consistent with these expectations, though further work is needed to assess the limits and scope of the generative approach.

8.6 Conceptual Implications

The generative framework shifts the foundational question from **“What mathematical structures describe physics?”** to **“What minimal structures might generate both mathematics and physics?”**

This reframing opens several possible directions for future research, including:

- identifying generative mechanisms in additional scientific domains,
- reinterpreting physical constants in terms of frequency-related quantities,
- exploring whether additional physical laws can be derived from oscillatory structure,
- examining polarity-based formulations of logic, computation, and information [1–4, 6–12, 13, 15–21].

These implications are exploratory, but they highlight the potential of generative approaches to inform foundational inquiry.

8.7 Why These Primitives?

A generative framework requires justification for the selection of its primitives. The present model adopts two: polarity as the minimal structural distinction and oscillation as the minimal dynamical expression of that distinction. This section explains why these primitives are selected, why they are minimally sufficient, and why alternative candidates—such as topology, information, symmetry, relation, or category structure—are not taken as foundational within this framework.

1. Why Polarity? (Minimal Structural Primitive)

Polarity is defined as a two-state involutive distinction. It is selected as the structural primitive because:

1. **Distinction is the first act of structure.** Any mathematical system—sets, logic, algebra, topology—begins with the ability to distinguish one state from another. Polarity is the minimal form of distinction.
2. **Polarity generates relation, order, and number.** From a two-state distinction, one can construct:
 - relations (same/different),
 - order (orientation between states),
 - number (iterated distinctions),
 - operations (concatenation of cycles).
3. **Polarity is strictly weaker than all alternatives.**
 - Set theory presupposes membership.
 - Logic presupposes truth values.
 - Category theory presupposes objects and arrows.
 - Topology presupposes open sets.
 - Information theory presupposes probability distributions. All of these require distinctions **already in place**. Polarity *is* the distinction.
4. **Polarity is ubiquitous across mathematics.** True/false, positive/negative, element/non-element, open/closed, convex/concave, symmetric/asymmetric—all are expressions of polarity.

Thus polarity is chosen because it is the **weakest possible structural assumption** capable of generating the structures mathematics requires.

2. Why Oscillation? (Minimal Dynamical Primitive)

Oscillation is selected as the dynamical primitive because:

1. **Oscillation is the temporal expression of polarity.** A two-state distinction that alternates in time produces periodic behavior. No additional assumptions are required.
2. **Oscillation generates frequency, phase, amplitude, and mode structure.** These are precisely the mathematical ingredients of wave equations.
3. **Oscillation is the simplest possible form of change.** Alternatives require more structure:
 - Differential equations presuppose calculus.
 - Hamiltonian mechanics presupposes phase space.
 - Field theory presupposes continuous fields.
 - Computation presupposes symbolic rules. Oscillation requires only alternation.
4. **Oscillation appears universally in physical law.** Foundational equations across classical mechanics, electromagnetism, quantum theory, relativity, and field theory admit frequency-mode solutions. This universality suggests oscillation is not incidental but generative.

Thus oscillation is chosen because it is the **minimal dynamical act** capable of producing the mathematical forms observed in physical law.

3. Why Not Other Primitives? (Exclusion of Alternatives)

Several alternative primitives have been proposed in foundational research. They are not adopted here because each presupposes more structure than polarity or oscillation.

Topology

Requires open sets, neighborhoods, and continuity. These cannot be defined without prior distinctions and relations.

Information

Requires probability distributions and entropy measures. These presuppose numerical structure and algebra.

Symmetry

Requires group structure. Groups require operations, identity, and inverses—none of which are primitive.

Relation

Requires a domain of objects to relate. Polarity generates relations; relations cannot generate polarity without assuming it.

Category Theory

Requires objects and morphisms. Both require distinctions and compositions—already downstream of polarity.

Computation

Requires symbolic states and transition rules. These presuppose distinctions, order, and operations.

In each case, the alternative primitive is **structurally heavier** than polarity or oscillation and therefore cannot serve as a minimal generative basis.

4. Minimal Sufficiency of the Chosen Primitives

Polarity and oscillation jointly satisfy the two requirements for a generative foundation:

1. **Structural sufficiency:** Polarity generates distinction $\rightarrow$ relation $\rightarrow$ order $\rightarrow$ number $\rightarrow$ operations $\rightarrow$ algebraic systems.
2. **Dynamical sufficiency:** Oscillation generates frequency $\rightarrow$ phase $\rightarrow$ amplitude $\rightarrow$ mode structure $\rightarrow$ wave equations.

3. No weaker pair of primitives is known to generate both mathematical and physical structure.

9. Limitations

Although the generative framework explored in this paper offers a minimal and coherent perspective on the origins of mathematical and physical structure, several limitations must be acknowledged. These limitations clarify the scope of the claims and identify areas where further development is needed [1–4, 6–12, 13, 15–21].

9.1 Formalization of the Generative Steps

The conceptual sequence from polarity to mathematical structure is outlined at a high level, but it has not yet been formalized within a rigorous axiomatic system. A complete formalization would require:

- explicit axioms for polarity,
- a precise definition of generative rules, and
- proofs establishing when and how specific mathematical structures arise.

The present work provides a conceptual foundation rather than a fully developed formal system [1–4, 6–12, 15–21].

9.2 Scope of the Frequency Hierarchy

The frequency hierarchy illustrates that many canonical equations in physics admit oscillatory or frequency-mode forms [1–4, 6–12, 13, 15–21]. However, the analysis focuses primarily on linear and widely used formulations. Extensions to:

- nonlinear systems,
- dissipative systems,
- strongly interacting quantum fields,
- emergent phenomena, and
- complex systems

may require additional investigation to determine whether oscillation remains generative or whether additional primitives are needed.

9.3 Dependence on Linearization

In several cases, oscillatory structure becomes apparent only after linearization. Newtonian mechanics reduces to harmonic motion under small displacements [1–3, 15], and gravitational waves arise from the weak-field limit of general relativity [10, 11, 21]. This raises the question of

whether oscillation is fundamental or whether it emerges from deeper nonlinear dynamics. Further work is needed to clarify the role of linearization in the generative interpretation.

9.4 Interpretation of Polarity in Physical Contexts

Polarity is introduced as a structural primitive for mathematics, but its physical interpretation is less direct. While oscillation can be viewed as a temporal expression of polarity, the mapping between mathematical polarity and physical systems may require additional clarification—particularly in contexts where physical states are continuous rather than discrete [1–4, 6–12, 15–21].

9.5 Empirical Testability

The framework is informed by the historical structure of physical equations, but it does not yet yield new quantitative predictions. A fully testable generative foundation would ideally produce:

- new constraints on physical laws,
- new relationships among physical constants, or
- new predictions about dynamical behavior.

The present work establishes a conceptual generative mechanism but does not yet derive novel empirical results [1–4, 6–12, 13, 15–21].

9.6 Compatibility with Existing Foundations

The framework is intended to complement, not replace, existing mathematical and physical foundations. Integrating polarity-based mathematics with set theory, category theory, or type theory may require additional interpretive layers. Similarly, embedding oscillation-based generative physics within established formalisms may require further conceptual development [1–4, 6–12, 13, 15–21].

9.7 Conceptual Minimality vs. Practical Complexity

Although polarity and oscillation are minimal primitives, the structures they aim to generate—mathematics and physics—are highly complex. The generative framework provides conceptual clarity but does not reduce the practical complexity of working within advanced mathematical or physical theories [1–4, 6–12, 13, 15–21].

9.8 Mathematical Formalization Roadmap

A complete generative foundation requires a rigorous formal system built on the primitives introduced in this work. While the present paper develops the conceptual structure of the polarity–oscillation framework, several steps are needed to formalize it mathematically.

Axiomatizing Polarity. A formal system would begin by specifying explicit axioms for polarity as a two-state involutive structure. These axioms would define the primitive distinction, the

involution operation, and the allowable transformations. Such a system would need to demonstrate how distinctions, relations, orderings, and numerical constructions arise from these axioms without presupposing additional primitives.

Formalizing Oscillation. Oscillation, interpreted as the temporal expression of polarity, would require a mathematical definition that connects the involutive structure to periodic behavior. This includes specifying the rules by which polarity generates temporal alternation, defining frequency as a derived quantity, and establishing the conditions under which oscillatory systems produce wave-equation forms.

Requirements for a Full Generative System. A complete formalization would need to show:

- how mathematical structures (numbers, operations, algebraic systems) arise uniquely from polarity,
- how physical structures (wave equations, dispersion relations, mode decompositions) arise from oscillation,
- how the two domains connect through frequency, and
- how the generative rules preserve consistency, completeness, and compatibility with existing mathematical and physical theories.

Developing this formal system is an essential next step. The present work provides the conceptual architecture, while a full axiomatization and quantitative development remain open directions for future research.

9.9 Final Remarks

A final limitation concerns the scope and intent of the framework itself. The polarity–oscillation model is presented as a conceptual foundation rather than a replacement for established mathematical or physical formalisms such as general relativity or quantum field theory. The framework outlines a possible generative structure beneath existing theories, but it does not yet provide the formal axiomatization, quantitative derivations, or predictive equations required for a full alternative formulation. Further work is needed to formalize the generative steps, develop explicit mathematical models, and derive quantitative predictions that can be compared directly with empirical data. These open questions define the next stage of development for the generative approach.

9.10 Related Work

The polarity–oscillation framework aligns conceptually with several foundational approaches that seek structural or relational origins for physical law. Wheeler’s pregeometry [28] proposed that spacetime geometry may emerge from deeper combinatorial or logical structures. Penrose’s twistor theory [29] introduced a relational representation of spacetime through complex geometric entities, emphasizing that geometry can arise from algebraic relations rather than

being fundamental. Barbour's shape dynamics [30] reinterprets gravitational theory in terms of relational configurations, removing absolute spacetime as a primitive. Rovelli's relational quantum mechanics [31] extends this principle to quantum theory, asserting that physical quantities exist only through relations between systems. Smolin's work on background independence and foundational structures [32] similarly explores how physical law may emerge from network-like or relational substrates.

The present framework shares with these approaches the goal of identifying minimal generative principles beneath established formalisms. However, it differs by proposing polarity and oscillation as the two primitives from which both mathematical and physical structures may arise. This situates the model within the broader tradition of relational and generative theories while maintaining its distinct focus on frequency as the structural bridge between mathematics and physics.

10. Predictions

A generative framework must yield consequences that can be evaluated against physical theory. The polarity–oscillation model developed in this work produces four classes of testable predictions derived from three assumptions: (1) polarity is the minimal structural distinction, (2) oscillation is the minimal dynamical expression of polarity, and (3) frequency links mathematical and physical structure. These predictions draw on established results across classical mechanics [1–3, 15], electromagnetism [4, 17], quantum theory [5–9, 13, 16, 18], relativity [10, 11, 21], wave-based geometric systems [23–27], and field theory [12, 19, 20].

10.1 Wave–Particle Duality as a Generative Prediction

Within the generative framework, any fundamental excitation of a field must exhibit both continuous oscillatory evolution and discrete particle-like interactions. Oscillation provides wave behavior through continuous temporal propagation [4, 17], while quantization produces discrete energy transfer events characteristic of particle-like detection [6, 13, 16, 18]. This predicts that wave–particle duality is not a duality but a structural consequence of quantized oscillatory modes, arising naturally whenever polarity is expressed dynamically and energy is exchanged in discrete units [1–4, 6–12, 15–21]. This prediction aligns with observed photon behavior and follows directly from the generative assumptions without additional postulates.

10.2 Oscillatory Compatibility of Future Physical Laws (Test Path 1)

Many foundational equations already take wave-like or frequency-mode forms, including Newtonian oscillators [1–3, 15], Maxwell's electromagnetic waves [4, 17], Schrödinger time evolution [6, 13, 16, 18], Klein–Gordon and Dirac plane waves [7–9], linearized gravitational waves [10, 11, 21], and quantum field modes [12, 19, 20].

Prediction: Any future fundamental physical equation must admit a frequency-mode representation.

Test Procedure: Given a proposed fundamental equation E , substitute a trial solution of the form

$$\psi(t) = e^{-i\omega t}$$

or a plane-wave ansatz. If no frequency-mode solution exists, the framework fails.

10.3 Frequency-Driven Geometric Differentiation (Test Path 2)

Wave-admitting systems show that increasing frequency produces higher symmetry orders, more intricate nodal patterns, and greater geometric differentiation. Examples include Chladni plates [23–27], membrane and cavity resonances [27], and optical/acoustic modes [14, 27].

Prediction: Increasing frequency must produce increasing geometric differentiation.

Test Procedure: Measure symmetry order, nodal complexity, and spatial differentiation across increasing frequencies in any wave-admitting system. If geometric differentiation does not scale monotonically with frequency, the framework fails.

10.4 Polarity-Based Mathematical Reducibility (Test Path 3)

Foundational mathematical structures used in physics—algebraic systems, vector spaces, Hilbert spaces, tensor calculus, and group theory—are built on distinctions, relations, orderings, and operations. These can be traced to polarity-based generative steps [1–4, 6–12, 15–21].

Prediction: Any mathematical framework fundamental to physics must be expressible as an elaboration of

$$\text{polarity} \rightarrow \text{relation} \rightarrow \text{order} \rightarrow \text{number} \rightarrow \text{operations}.$$

Test Procedure: For any mathematical structure claimed to be fundamental, demonstrate whether it can be reduced to polarity-based generative steps. If a required structure cannot be reduced, the framework fails.

10.5 Universality of Frequency in Dynamical Systems (Test Path 4)

Frequency appears universally across classical mechanics [1–3, 15], electromagnetism [4, 17], quantum mechanics [5–9, 13, 16, 18], relativistic field theory [7–9], gravitational waves [10, 11, 21], and quantum field theory [12, 19, 20].

Prediction: Any lawful dynamical evolution must be representable in frequency-mode form.

Test Procedure: Analyze new dynamical systems for oscillatory or mode-based decompositions. If a lawful dynamical system cannot be expressed in frequency form, the framework fails.

10.6 Consolidated Testable Consequences

The four predictions above yield explicit, executable tests:

- **Oscillatory compatibility test:** Any new physical equation must admit a frequency-mode solution.
- **Symmetry–frequency scaling test:** Geometric differentiation must increase with frequency [23–27].
- **Mathematical reducibility test:** Mathematical structures must be derivable from polarity-based distinctions [1–4, 6–12, 15–21].
- **Frequency universality test:** Any dynamical system must exhibit oscillatory or frequency-mode behavior.

10.7 Potential Falsifiers

The framework is falsified if any of the following are observed:

1. A fundamental physical equation that cannot be expressed in oscillatory or frequency-mode form.
2. A mathematical structure essential to physics that cannot be reduced to polarity-based generative steps.
3. A wave-admitting system in which geometric differentiation does **not** scale with frequency [23–27].
4. A dynamical system that exhibits lawful evolution but **no** oscillatory or frequency-mode representation.

Any of these outcomes would contradict the generative assumptions and invalidate the framework.

10.8 Quantitative Prediction

In any wave-admitting geometric system, the node-density $D(f)$ of the resulting pattern must scale monotonically with driving frequency f . For thin-plate systems governed by the biharmonic equation, the framework predicts an approximate power-law relationship:

$$D(f) \propto f^{1/2}.$$

Testability: Measure node-density across increasing frequencies in a vibrating-plate or membrane system. If the empirical scaling exponent deviates significantly from 1/2, the generative framework is falsified.

10.9 Quantitative Prediction: Symmetry–Frequency Scaling

In any two-dimensional wave-admitting system with fixed boundary conditions, the rotational symmetry order $S(f)$ of the standing-wave pattern must scale with driving frequency according to:

$$S(f) \approx \alpha f^{1/2},$$

where α is a system-dependent constant determined by plate geometry and boundary constraints.

Testability: Measure the rotational symmetry order of nodal patterns across increasing frequencies. If the empirical scaling exponent differs significantly from $1/2$, the generative framework is falsified.

10.10 Quantitative Prediction: Frequency–Energy Scaling in Oscillatory Systems

In any system where oscillation serves as the fundamental dynamical act, the energy associated with a stable mode must scale linearly with its characteristic frequency. This follows from the generative assumption that quantization arises from discrete oscillatory cycles, yielding the relationship:

$$E(f) = \beta f,$$

where β is a system-dependent proportionality constant determined by boundary conditions and physical parameters. This prediction mirrors the well-established linear frequency–energy relationship observed in quantized electromagnetic modes [4, 17] and quantum mechanical energy levels [6, 13, 16, 18].

Testability: Measure the energy of stable oscillatory modes across increasing frequencies in any quantized or semi-quantized system (optical cavities, acoustic resonators, or mechanical oscillators). If the empirical energy–frequency relationship deviates significantly from linearity, the generative framework is falsified.

10.11 Quantitative Prediction: Mode Density Scaling in Bounded Oscillatory Systems

For any finite system supporting standing wave modes, the number of admissible modes below a given frequency must scale quadratically with frequency in two dimensions and cubically in three dimensions. This follows from the generative assumption that geometric differentiation increases with frequency and that mode structure arises from discrete oscillatory configurations [23–27]. In two dimensions, the prediction takes the form:

$$N(f) \propto f^2,$$

and in three dimensions:

$$N(f) \propto f^3.$$

These relationships are consistent with known results for membrane, cavity, and field-theoretic mode densities [12, 19, 20].

Testability: Count the number of standing wave modes below frequency f in a bounded system (e.g., a drumhead, optical cavity, or acoustic chamber). If the observed mode density scaling exponent differs significantly from 2 (in 2D) or 3 (in 3D), the generative framework is falsified.

11. Conclusion

This work develops a generative framework in which polarity serves as the minimal structural distinction, oscillation serves as the minimal dynamical expression of that distinction, and frequency provides the common descriptor linking mathematical and physical structure. By treating polarity and oscillation as primitives rather than derived constructs, the framework offers a unified basis for understanding why wave equations, mode structures, and frequency-based formulations appear so widely across physical law.

The mathematical development shows how polarity generates relations, order, number, and operations, providing a structural pathway from minimal distinction to the mathematical tools used throughout physics. The physical development demonstrates that oscillation naturally yields the dynamical forms observed in classical mechanics, electromagnetism, quantum theory, relativity, and field theory. Across these domains, frequency emerges as a universal descriptor of lawful change.

The predictions derived from the framework identify clear avenues for evaluation. These include the requirement that future physical laws admit frequency-mode representations, the expectation that geometric differentiation scales with frequency in wave-admitting systems, the reducibility of fundamental mathematical structures to polarity-based generative steps, and the universality of oscillatory behavior in dynamical systems. Each prediction yields explicit test paths and falsifiers, ensuring that the framework remains empirically accountable.

Taken together, the results suggest that polarity and oscillation may provide a minimal generative basis for both mathematical structure and physical law. The framework does not replace existing theories; rather, it identifies the structural and dynamical primitives that underlie them. Future work may extend this approach by examining additional domains where frequency-mode formulations appear, exploring the implications for unification, and refining the testable consequences outlined in this manuscript.

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